



MAPS 4

Best Practices

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Version	Date	Author	Comment	Approval
1.0	02.02.2018	D. Gibson	Original	
2.0	25.05.2020	J. Duncan	Update formatting	
3.0	29.06.2020	J. Duncan	Update contents, flow and Cover	
3.0	03.07.2020	D. Gibson	Review	
4.0	14.08.2020	J. Duncan	Reformat	
5.0	17.06.2021	J. Duncan	Finalise	C. Swanepoel
6.0	18.06.2021	J. Duncan	Minor corrections	
7.0	30.09.2021	J. Duncan	QMS	

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1. Introduction

This document serves as a basis against which any SCADA project can be specified and designed. Whilst it covers all the necessary standards for implementation it is intended to stimulate forethought and discussion with respect to the customer requirements and the process environment and how these will impact on the architecture, configuration, deployment of a MAPS SCADA project.

2. Pre-configuration

Before work can begin on the SCADA project, the engineer will need to understand the user's requirements.

To start with, the engineer will need the following:

2.1. Documentation

2.1.1. Input output schedule (I/O Schedule)

An I/O schedule is a spreadsheet that lists all PLC inputs and outputs that are required in the SCADA. The I/O Schedule should contain the following columns:

- PLC memory address
- PLC data type
- Agent name
- Agent description
- Agent type
- Device minimum and maximum values
- Engineering minimum and maximum values
- Alarm set points
- Is data logging required
- Alarm Type

2.1.2. User Requirements Specification (URS)

A URS is a description of the SCADA system to be developed and is created with the end user. It lays out functional and non-functional requirements and may include a set of use cases that describe user interactions that the software must provide.

2.1.3. Functional Design Specification (FDS)

An FDS is a formal document used to describe in detail the SCADA's intended capabilities, appearance, and interactions with users. The functional specification is a kind of guideline and continuing reference point for the SCADA engineer as he develops the operations and activities that the system must be able to perform and define how to meet the User Requirement (see User Requirements Specification).

2.2. System Architecture

2.2.1. Distributed System

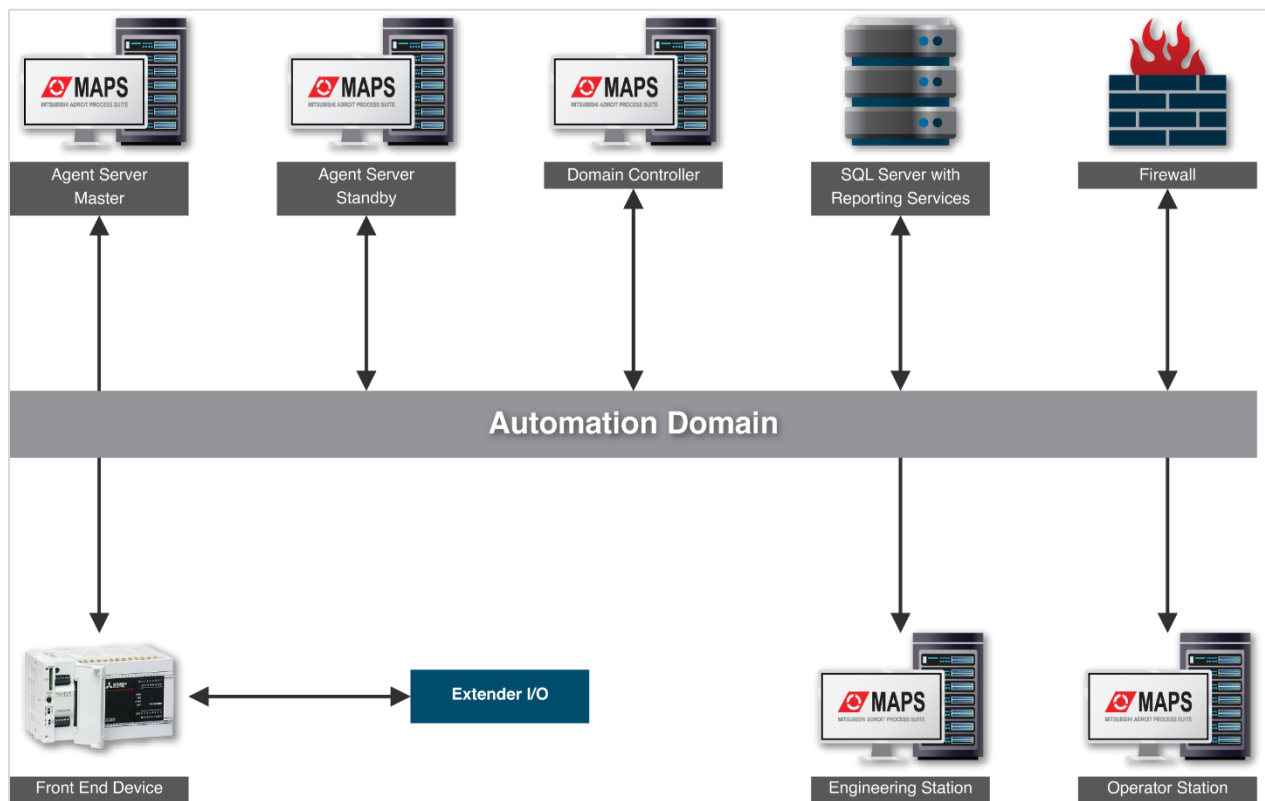


Figure 1 – Distributed System Architecture

2.2.2. Recommended Hardware & Software

Please refer to the Hardware & Software Specification document available from our website.

<https://www.mapsscada.com/downloadsacc/>

2.2.3. Firewall

The Automation SCADA network should be isolated from any external network traffic. If access is required a hardware firewall is highly recommended.

2.3. Standalone System

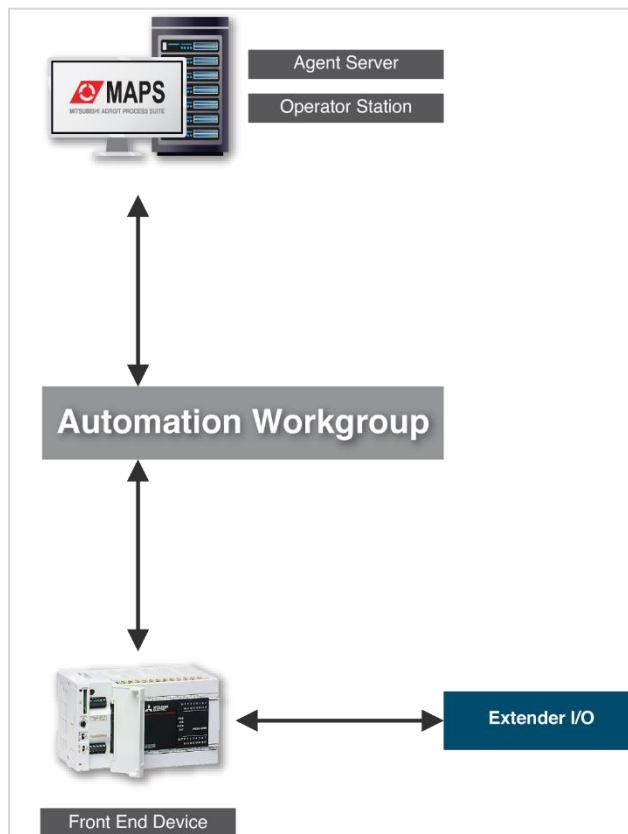


Figure 2 - Standalone System Architecture

2.3.1. Recommended Hardware & Software

Please refer to the Hardware & Software Specification document available from our website. Vehicle

<https://www.mapsscada.com/downloadsacc/>

2.4. Network requirements

MAPS SCADA's minimum requirement for a local area network (LAN), is as follows:

- A minimum of 100 MB ethernet backbone. We recommend a 1gb ethernet connection.
- All connected devices (including PC network cards and switch ports) to be configured for full duplex operation.
- All connected PC's that have MAPS SCADA (or MAPS SCADA components) installed to be able to see remote MAPS SCADA PC shares.
- Proper separation of PLC and control networks. While this is not a requirement for MAPS SCADA, this has historically been an important issue on PLC networks whose ethernet cards can only handle 10MB. If such a 10 MB PLC network is flooded with control (or office) network traffic, PLC communications will be compromised. This requires an important understanding (and measurement of) PLC network traffic to ascertain whether "cross-talk" might be an issue.
- If a domain name server (DNS), is not available. Then the file C:\Windows\System32\drivers\etc\hosts will need to be edited on all client and server machines to include all the IP addresses and PC/server names. [The knowledge base article.](#)

Complete these “basic” network tests:

- Ping tests of 65500 byte packets should occur without any failures using the ping command as follows: “ping PC/Server Name -t -l 65500” The PC/Server name must resolve on the network.
- Browsing between remote UI and server’s file shares via Windows Explorer’s Network Folder should not fail or prompted for user authentication.
- The copying of a large file to and from remote PC’s should register a full 100 MB usage on the Task Manager’s network graph.
- Disconnect the PC from control network and ensure that remote PC shares cannot be accessed via the PLC network connection.

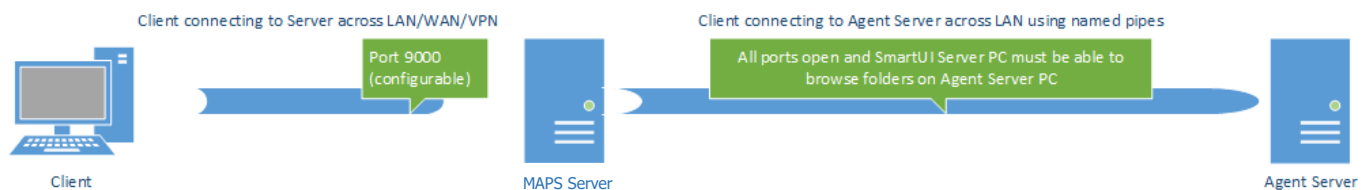
Important potential points of failure:

- Network card duplex and speed configuration should be checked. “Auto” settings are not suitable as Windows may choose to select half duplex modes. It is preferable to enforce full duplex selection. Disable any power management in the properties of the network card.
- Network switch ports not configured (or running) for full duplex (historically, this has been a prevalent issue across many larger installations when receiving and transmitting data).
- Poorly maintained network infrastructure like incorrectly crimped RJ45’s or damaged flyleads.
- Use of incorrect Ethernet cable specification for the type of network.
- Incorrectly set share permissions for access by remote PC’s.
- Incorrect separation of control and PLC networks. It helps to remove the “Client for Microsoft Networks” bindings on PLC adaptor cards.

2.5. Network Configuration

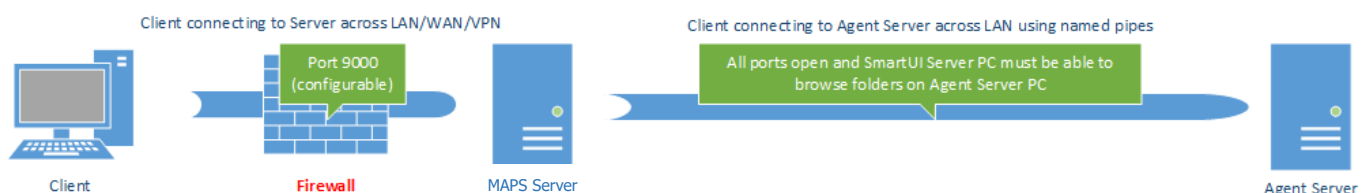
There are several different ways to configure the communication/network infrastructure for the MAPS SCADA product. Four of the main default network layouts are:

- Connecting directly



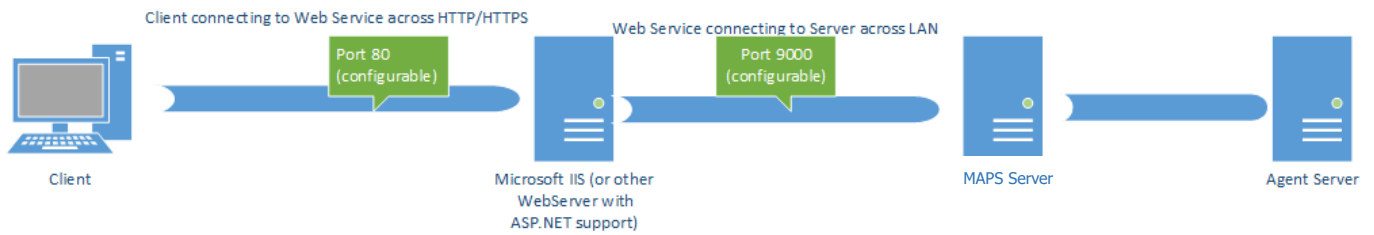
This network architecture is one of the most widely used, and there are no firewalls to configure.

- Placing a firewall before the SmartUI Server



The firewall needs to allow port 9000 communication from the Client to the SmartUI Server

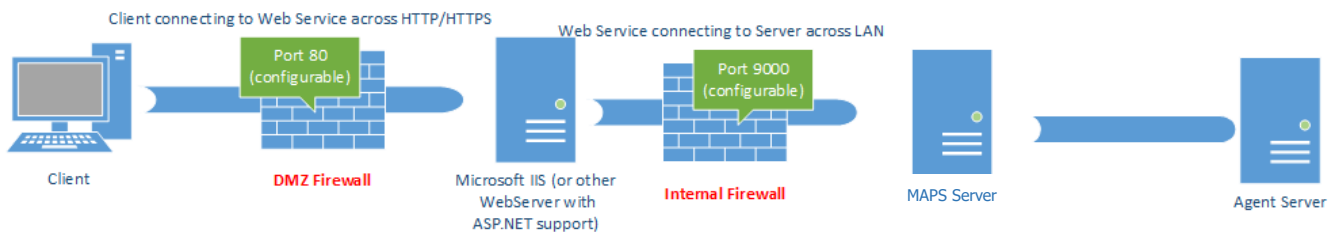
- Connecting via HTTP



This configuration only requires port 80 to be open to external access and all client communication will utilise HTTP only.

(Please note that this is a poll only configuration, so turnaround times are slightly slower than scenario 1 or 2)

- Setting up a DMZ



(Note: The DMZ configuration can also be used in a LAN/WAN environment. The image below explains it in terms of a HTTP/Web environment)

2.6. Licensing

Typically, the Agent Server is licensed via a HASP software protection key, which is inserted into a USB port of the computer on which the Agent Server is running. This HASP is then programmed in accordance with your specific license agreement.

In situations where it is not possible to use a physical HASP, it is also possible to license MAPS SCADA via a software license. Software licensing is done against the server's hardware so any changes to the physical hardware will invalidate the soft license.

2.6.1. Version

HASP's are licensed against versions of MAPS SCADA.

2.6.2. Scan points

When a PLC address is scanned a Scan point is consumed. For every DBAccess; script; and SNMP agent a scan points is also consumed.

2.6.3. Remote Operator/UI

A remote node licence is required for every client connection.

2.6.4. OEM

Additional components are licensed against the OEM code of the HASP. e.g. Alarm management; Max Demand; Siemens Ethernet driver; DNP 3.0 driver.

2.6.5. Generating licensing

The Adtech Licensing Utility allows you to get, view and/or update your MAPS SCADA product license. Use the Adtech Licensing Utility to generate a licensing request by launching the application via the MAPS SCADA Utility launcher application. By clicking on the icon:



Then selecting the Adtech Licensing Utility.

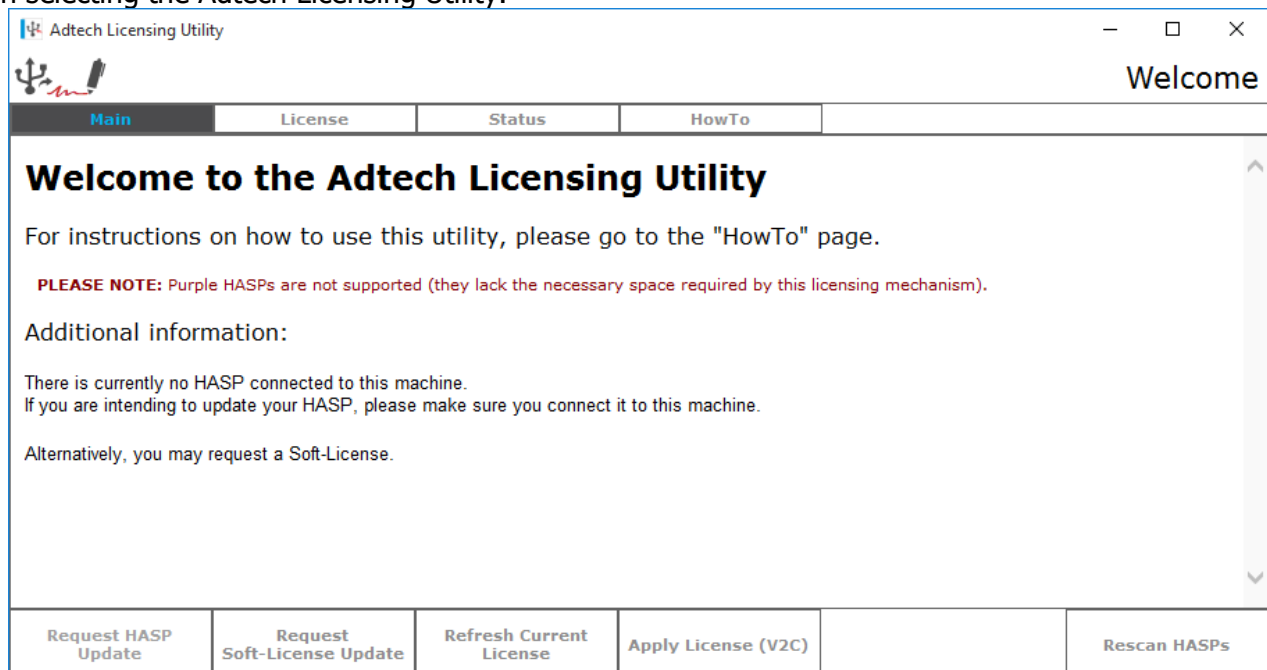


Figure 3 - Adtech Licensing Utility

Select either the HASP or Soft-License update to generate the c2v file. Once the file has been generated it can be emailed to MAPS SCADA with the licensing requirements.

2.7. Security

2.7.1. Security Groups

- **Default Logon** - This user will be used to auto login to the MAPS SCADA server. Once the user has logged in, they will be prompted to fill in their user names and passwords. This user has no rights on the system other than navigation. The user name for the default login is Defaultop.
- **Operator** - There will be one operator group. An Operator will manage the day to day running of the site's normal alarms and events.
- **Engineer** - There will be one engineer group. The engineers will be responsible for adding; commissioning and maintaining of sites.
- **Application Admin** - The application admin group will contain user that design and engineer projects. This group will have full rights across the entire solution.
- **Domain Administrators** - This group will have full administrator rights on the domain but will have no rights in the MAPS SCADA application.

2.7.2. Security Group Rights

Actions	User Groups				
	Default Login	Operator	Engineers	Application Admin	Domain Administrators
View and navigation	✓	✓	✓	✓	X
Acknowledge Alarms	X	✓	✓	✓	X
Write to Set Points	X	X	✓	✓	X
Reset Devices	X	✓	✓	✓	X
View/Query Maintenance Data	X	✓	✓	✓	X
Update/Change Maintenance Data	X	✓	✓	✓	X
View Device Home Form	✓	✓	✓	✓	X
View Device Setup Form	X	X	✓	✓	X
Commissioning and Operation	X	X	✓	X	X
MAPS SCADA Designer	X	X	✓	✓	X
Commissioning and Operation	X	X	✓	X	X
MAPS SCADA Designer	X	X	✓	✓	X

3. Context Model

Context model defines how the data will be structured and maintained. It aims to produce a formal or semi-formal description of the context information that is present in a context-aware system. In other words, the context is the elements for the system, and a model provides the mathematical interface and a behavioural description of the surrounding environment.

A key role of context model is to simplify and introduce greater structure into the SCADA solution. The context model will also help apply structure to any reports required. MAPS SCADA uses the S88/S95 Standard as a starting point.

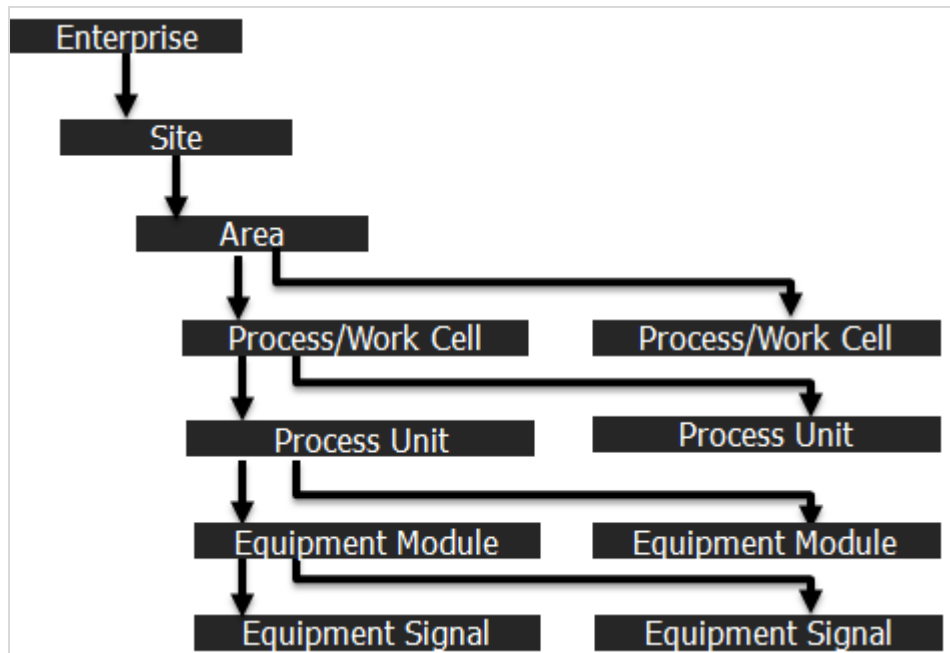


Figure 4 - S88/S95 context model

Using the one to many S88 structure above the engineer can define the context as below.

S88 Name	Adroit Context	Specified Name	Abbreviation	Related To
Enterprise	Configuration	Adroit Tech.	ADR	
Site	Agent Server	AS 01	AS_01	ADR
Area		Production	PRD	AS01
Area		Quality Control	QCL	AS01
Process Cell	Device	PLC-A	PLCA	PRD
Process Cell	Device	PLC-B	PLCB	QCL
Unit		Booster Pump	BST_PMP	PRD
Equipment		Pump A	PMP_A	BST_PMP
Signal		Trip	TRP	PMP_A

Another way to define the context

Enterprise – Adroit Tech.	
Site – AS 01	
Area – Production	Area – Quality Control
Process Cell – PLC-A	Process Cell – PLC-B
Unit – Booster Pump	
Equipment – Pump A	
Signal – Trip	

4. Agent Server Configuration

4.1. Agent Server name

The Server Project Name and Agent Server name are defined in the Context Model. Typically this means that the project name is the same as the Enterprise and the agent server name is the same as the site.

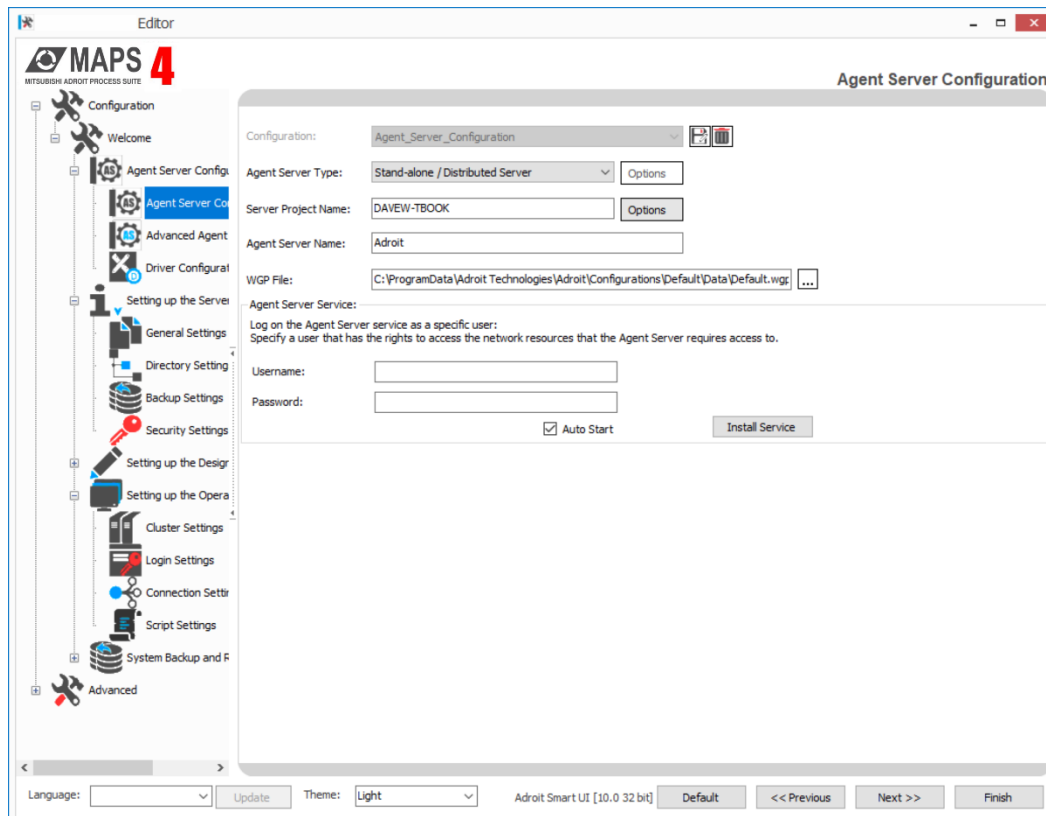


Figure 5 - S88/S95 context model

4.2. Naming Convention

Context model defines the naming convention.

4.2.1. Tag

VVV_WWW_XXX_YYY_ZZZ_##

Symbol	Description*	Abbreviation
VVV	Production	PRO
WWW	PLC-A	PLA
XXX	Booster Pump	BSP
YYY	Pump A	PPA
ZZZ	Trip	TRP
##	Module number	01

*As per the Context Model's Specified Name

Tag name = PRO_PLA_BSP_PPA_TRP_01

4.2.2. Device

VVV_WWW_##

Symbol	Description*	Abbreviation
VVV	Production	PRO
WWW	PLC-A	PLA
##	An incremental number to support multiple devices	00

*As per the Context Model's Specified Name
Device Name = PRO_PLA_00

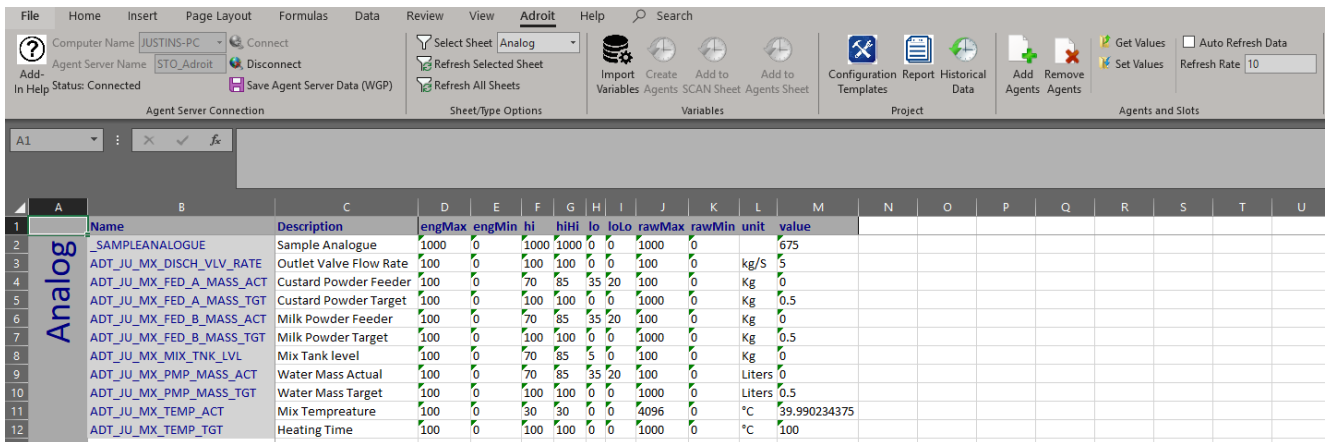
4.3. Bulk Configuration

Configuration data can be converted from Agent Server load files (*.WGP) into comma separated variable files (*.CSV) by using the Export command from the File menu or from the configurator's Edit dialog. Third party applications can then be used for off-line editing and modification prior to re-Import into an Agent Server load file.

MAPS SCADA also offers an Excel Add-in Tool that assists with the bulk configuration of tags. The MAPS SCADA Add-in for MS Excel is a bulk offline configuration tool for your MAPS SCADA agent database file, which is primarily intended for advanced users.

However, in addition to being a bulk SCADA configuration tool, this tool allows you to:

- Configure the security and auditing properties of the required agents and their slots.
- Create report/s that display MAPS SCADA Agent Values.
- View historical data of a logged tag.



	Name	Description	engMax	engMin	hi	hiHi	lo	loLo	rawMax	rawMin	unit	value
1	_SAMPLEANALOGUE	Sample Analogue	1000	0	1000	1000	0	0	1000	0		675
2	ADT_JU_MX_DISCH_VLV_RATE	Outlet Valve Flow Rate	100	0	100	100	0	0	100	0	kg/S	5
3	ADT_JU_MX_FED_A_MASS_ACT	Custard Powder Feeder	100	0	70	85	35	20	100	0	Kg	0
4	ADT_JU_MX_FED_A_MASS_TGT	Custard Powder Target	100	0	100	100	0	0	1000	0	Kg	0.5
5	ADT_JU_MX_FED_B_MASS_ACT	Milk Powder Feeder	100	0	70	85	35	20	100	0	Kg	0
6	ADT_JU_MX_FED_B_MASS_TGT	Milk Powder Target	100	0	100	100	0	0	1000	0	Kg	0.5
7	ADT_JU_MX_MIX_TNK_LVL	Mix Tank level	100	0	70	85	5	0	100	0	Kg	0
8	ADT_JU_MX_PMP_MASS_ACT	Water Mass Actual	100	0	70	85	35	20	100	0	Liters	0
9	ADT_JU_MX_PMP_MASS_TGT	Water Mass Target	100	0	100	100	0	0	1000	0	Liters	0.5
10	ADT_JU_MX_TEMP_ACT	Mix Temperature	100	0	30	30	0	0	4096	0	°C	39.990234375
11	ADT_JU_MX_TEMP_TGT	Heating Time	100	0	100	100	0	0	1000	0	°C	100

Figure 6 Bulk Configuration

4.4. Scanning

4.4.1. Scanning sub-system

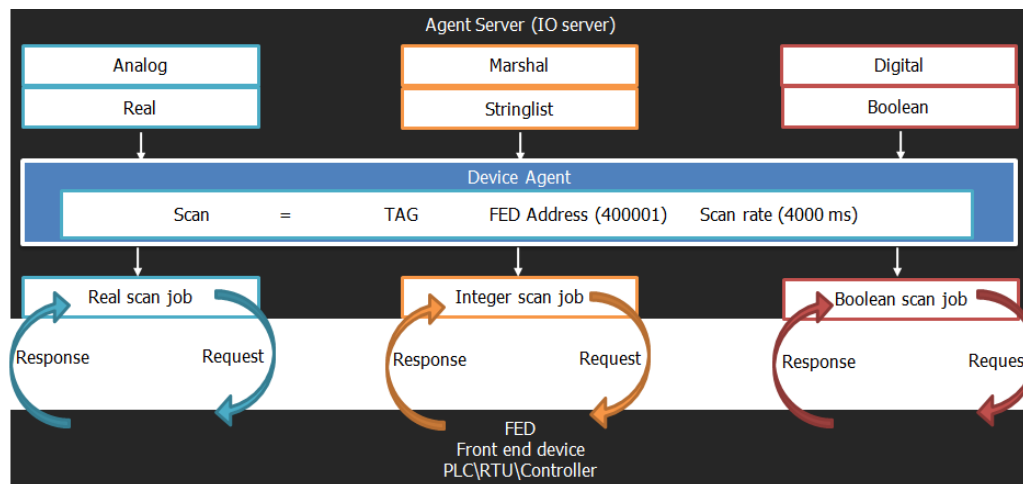


Figure 7 - Scanning subsystem - Solicited/Polled Driver Architecture

4.4.2. Factors affecting scanning performance

- Bandwidth and other medium-specific issues.
- The number of devices per communications channel.
- Total number of transactions required to refresh each device on the network – contiguous data packing
- The frequency of transactions (Scan Rate).

4.4.3. Calculating the scan period.

It is very easy to oversubscribe (scan) to data, there are limitations to any communications medium – the document will assume serial-based communications, as it is the one most easily overestimated. A few assumptions will have to be made:

- The total number of Outstations on the network.
- The total number of I/O scan tasks per outstation (approx.) and their size in characters (approx.).
- Approximate/Average Turnaround Time for a transaction. (See diagram)
- Network characteristics such as baud rates/bandwidth etc.
- Assume for the ideal case, I.e. No errors, retries, pre-transmission delays, latency delays etc.

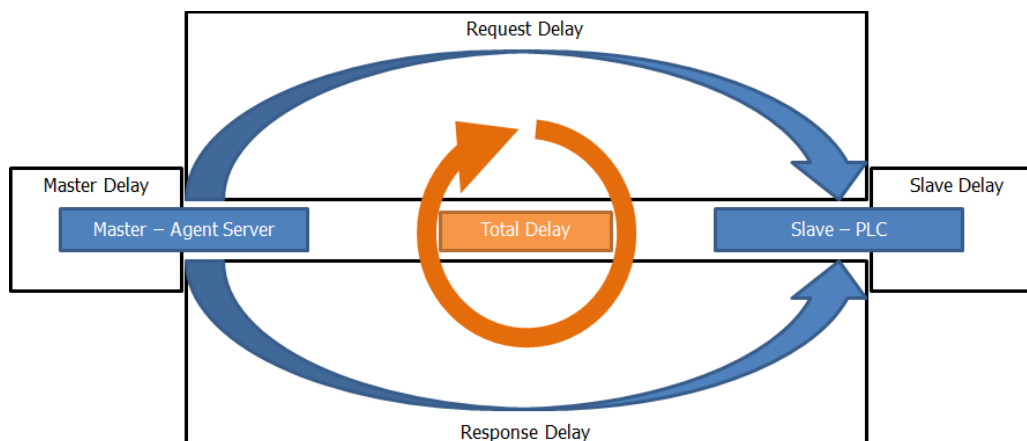


Figure 8 - Scanning transaction time

Example - Serial communication calculation

9600 baud rate, 60 outstations, 2 (100 chars) Transactions per outstation per refresh, and a total turnaround of 300 milliseconds per transaction. The 100 chars above includes the request and response and any acknowledgements.

9600 allows 1 bit every 0.0001 seconds, assuming 1 start bit and 2 stop bits, 11 bits per character, means ~872 characters per second on the channel if the channel runs at maximum. $60 \text{ (outstations)} \times 2 \text{ (transactions)} \times 100 \text{ (chars)} = 12000 \text{ chars to refresh the entire network. } 12000 \text{ (chars in total)} / 872 \text{ (channel config max)} = 13.72 \text{ Seconds to refresh the entire network at maximum channel capacity.}$

100 characters per transaction roughly use 114 milliseconds network time but there will be a turnaround time conservatively assumed above as 300 milliseconds thus: $300 * 120 \text{ (total network transactions)} = 36 \text{ seconds to refresh the entire network.}$

Now this means if every transaction is an equal priority and assuming no errors, delays, or pre-transmission delays etc. no scan rate should exceed 36 seconds.

Implement the following solutions, to shorten the scan rate:

- Add additional 3 channels, the network update time will now be 9 Seconds.
- Prioritise scan tasks, use a slower scan rate for data that updates slower in the field, e.g. reservoir levels (e.g. Integrity polls), high priority data faster (e.g. Digital status indications).
- Use an exception based protocol where the fastest way to get a change of state data is via Unsolicited transactions. Then the other data is divided up into priorities just like option 2 above.

Every effort is to be made when drawing up the I/O schedule of the PLC/SCADA and to define logical and contiguous data areas to be scanned from Devices. This makes for efficient, reliable and fast scanning.

Take note of the driver documentation that will assist in creating an optimum scanning (and hence performance) of the MAPS SCADA scanning system.

4.5. Alarming

The solution is to be configured in accordance with best practices. In principle, it has to be identified what constitutes an alarm and what constitutes an event and the two kept separate. Applying the definition that an "Alarm is an event that requires an operator to acknowledge and take some action", thereby not cluttering up the alarm viewers with unnecessary data.

To start it is recommended that all the defined alarms be broken into three categories or tiers. Each tier can be assigned a route in the Alarm agent and an outcome using the associated agents.

For example:

- **Events – Route 2**
 - Global Alarm list – An alarmlist agent that is assigned to all routes.
 - Alarm Management - The Alarm Management agent creates and maintains a SQL database for long-term alarm reporting.
- **Non-critical Alarms – Route 3**
 - Global Alarm list
 - Non-critical Alarm list - An alarmlist agent that is assigned to route 3, These alarms require the Operator to acknowledge them.
 - Alarm management
- **Critical Alarms – Route 4**
 - Global Alarm list
 - Non-critical Alarm list - An alarmlist agent that is assigned to route 4, these alarms require the Operator to acknowledge them.
 - SMS – A notify agent that sends the alarm to a predefined cell number.
 - Alarm management

4.6. Data Logging

4.6.1. Logging for Trends

The MAPS SCADA data logging sub-system is used to store/log historical data for trending purpose. The data log should be configured as follows.

- Data source – A data log agent logs the data to a lgd file, all required signals belonging to an equipment model should be logged to the same lgd file. The lgd file name should be made up using the context model.
- Using the length and deadband time the MAPS SCADA agent server will calculate the required size of the lgd file. The lgd file can be no larger than 2 GB so care should be shown when specifying the length and deadband time.
- Length – The length of time required to store the data.
- Deadbands Time – The time or sample rate of a datalog should be faster than the scan rate. if your scan rate is 1000 ms then the datalog time should be 900 ms.

4.6.2. Logging for Reports

The DBlog agent can be used to log data to SQL database. There should be two DBlog agents:

- Digital DBlog – All digital agents and marshal agent bits should be added to this DBlog agent. The logging method should be set to Log all on change.
- Analogue DBlog – All analog agents should be added to this DBlog agent. The logging method should be set to Log all on interval. The logging interval is usually user defined; however, a period of 5 minutes is customary.

5. Design Principles

Much has been learned over the years regarding good and bad HMI design practices. Having learned through trial and error, the good is retained and the bad disregarded. Below are three primary principles which need to be implemented for an efficient HMI design.

5.1. Three Primary Principles

5.1.1. Clarity

- Graphics are easy to read and intuitively understandable.
- Graphics show the process state and conditions clearly.
- Graphic elements used to manipulate the process are clearly distinguishable and consistently implemented.
- Graphics do not contain unnecessary details and clutter.
- Graphics convey relevant information, not just data.
- Information has prominence based on relative importance.
- Alarms and indications of abnormal situations are clear; prominent; and consistently distinguishable.

5.1.2. Consistency

- Graphics functions are standardised, intuitive, straightforward and involve minimum keystrokes or pointer manipulations.
- The HMI is set up for navigation in a logical, hierarchical and performance-oriented manner.

5.1.3. Feedback

- Graphic elements and objects (wizards/controls) must behave and function consistently in all graphics and all situations.
- Important actions with significant consequences will have confirmation mechanisms to avoid inadvertent activation.
- Design principles will be used to minimize user fatigue, since operations personnel use the graphics constantly.

5.2. Graphic Hierarchy

High-performance graphics are designed in a hierarchy for progressive disclosure of further details.

- Level 1 – Process Area Overview.
- Level 2 – Process Unit control.
- Level 3 – Process Unit Detail.
- Level 4 – Process Unit Support and Diagnostic Displays

6. SmartUI Server Configuration

6.1. Configuration

The new configuration for the SmartUI should be named using the Context Model. The name could be the same as the project name.

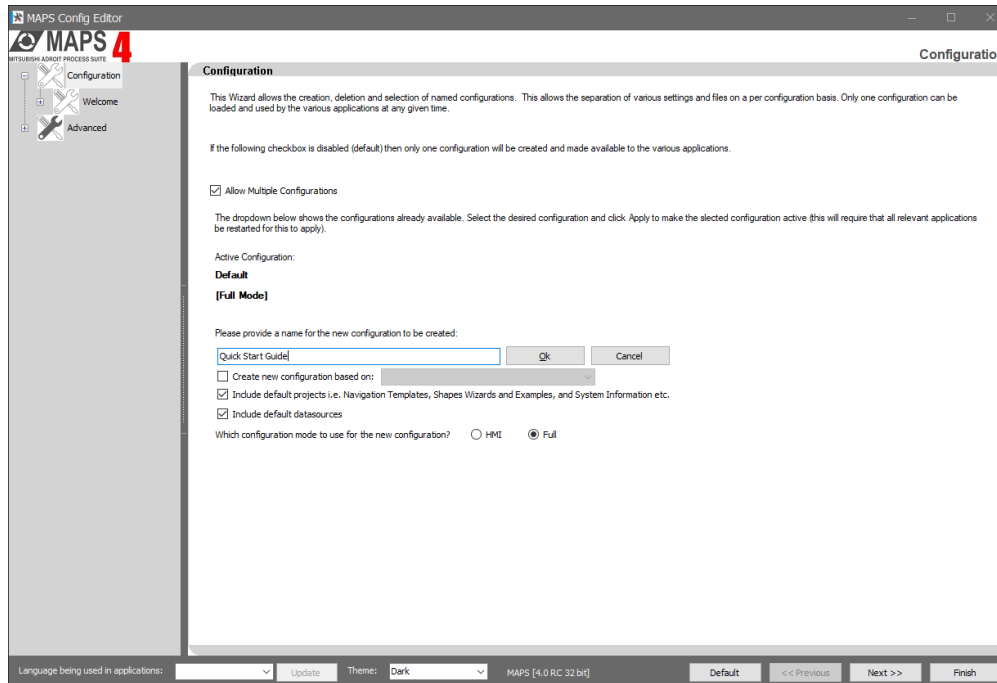


Figure 9 MAPS SCADA SmartUI Editor

6.2. Datasources

When creating a datasource the name of the datasource should reflect what it is connecting too. For example, the name for the MAPS SCADA datasource should be the same as the agent server's name and the OLE DB datasource name should be the database's name that it connects too.

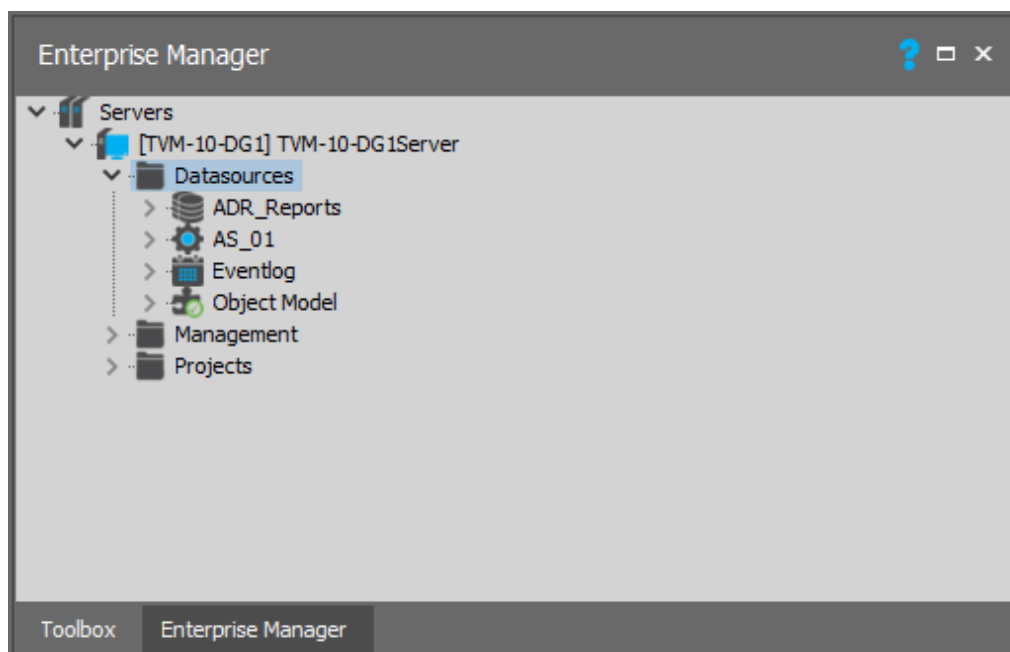


Figure 10 - SmartUI data sources

6.3. Project

6.3.1. Default Settings

Screen Size

Machine	Screen Size pixels	Aspect ratio
Local View stations in control room	1920 x 1080 (HDMI)	16:9
Remote clients (example laptop)	1920 x 1080 (HDMI)	16:9

Fonts

Topic	Font	Size
Default	Tahoma	10pt
Main Heading	Tahoma	16pt (Bold)
Sub Heading	Tahoma	12pt

Colours

Default	
Back colour	White
Fore colour	Black

Display Value	
Back colour	White
Fore colour	Black

Set Point Value	
Back colour	White
Fore colour	Blue

Normal Operation	
Back colour	Light steel blue

Warning	
Back colour	Amber

Fault/Alarm	
Back colour	Red

Stopped	
Back colour	White
Fore colour	Black

Started/Running/Active	
Back colour	Dim Grey
Fore colour	White

6.3.2. Project Structure

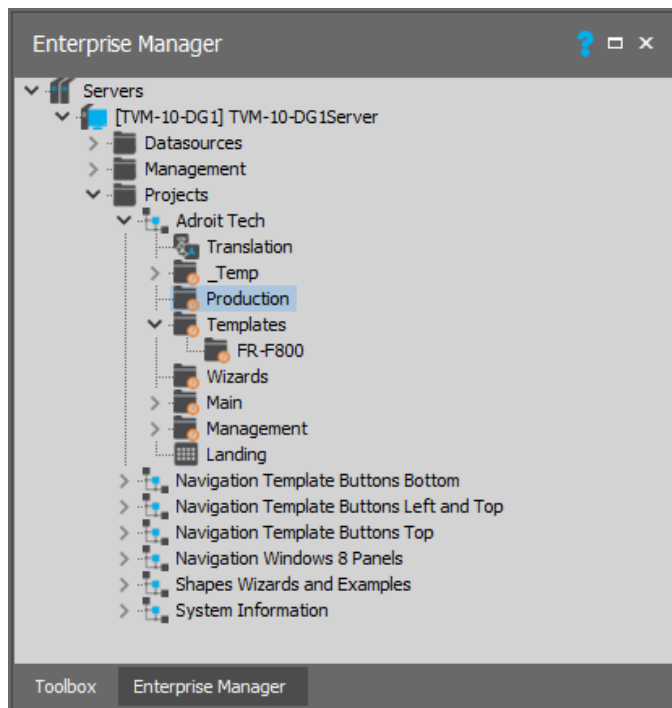
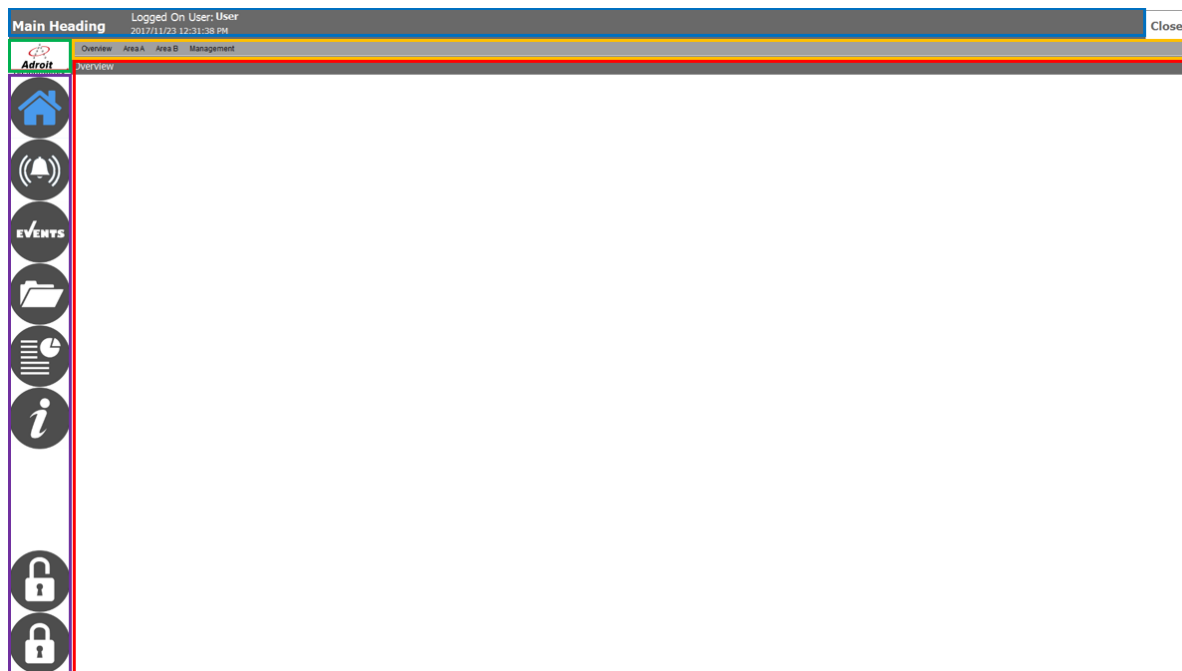


Figure 11 - SmartUI project structure

- Project Name – Adroit Tech.
- Landing: Main navigation graphic form.
- Folder – Main: Contains all graphic forms that are used across the entire projects.
- Folder – Production: Create a folder for each area of the context model, contains graphic forms that are associated with the area.
- Folder – Wizards: Contains all wizards.
- Folder – Templates: Contains sub-folders for each equipment in the context model.
 - a. Sub-folder – FR-F800: Contains templates (faceplate, settings, alarms and events) for associated VFD.

6.3.3. Screen Layout and Navigation

When creating a user interface a basic rule should be kept in mind, Left to right top to bottom. This means that typically the operator will always look to the top left of the screen when operating the plant.



- **Main heading and system information** – This area should display all system overview information, the simplest way to think of this area is a heads-up display to alert operators to the overall health of the system.
- Corporate logo - Optional
- **Main Navigation** – Contains the navigation that for example will allow the operator to load

Icon	Description
	SCADA Overview
	Global and historical alarms
	System events
	SCADA and project documentation
	Reports
	System information and management
	Logon with new user
	Logoff current user and logon with default user

- **Sub Navigation** – Sub menu is dependent on main navigation selection.
- **Graphic form loading area** – template graphic object that allows the operator to load the required graphic form.